Unit 2: Supply

Objectives

After studying this unit, the student should be able to:

- understand the concept of supply curves
- define supply and its determinants
- explain the Law of Supply
- identify differences between Individual and Market Supply Curve

Supply

Supply of a commodity means quantity of the commodity which a firm is willing to sell at a given price during a particular time. Like demand, supply definition is complete when it has the following elements:

(i) Quantity of a commodity that the producer is willing to offer for sale,
(ii) Price of the commodity, and
(iii) Time during which the quantity is offered for sale.

Example: Firm A supplies 50 kg of wheat at price of ₹10 per kg in a month is a statement of supply.

Supply function

Supply function is a functional relationship between quantity supplied of a commodity and factors affecting it. The supply function can be written as:

$$\mathbf{S_X = f\ (P_X, P_Z, T, C, G_P)}$$

where,

S_X = Supply of commodity X

f = function of

P_X = Price of commodity X

P_Z = Price of related good, Z

T = Technological changes

C = Cost of production or price of inputs

G_P = Government policy or excise tax rate.

Factors Affecting Supply of a Commodity

a) Price of the Commodity

At a higher price, producer offers more quantity of the commodity for sale and at a lower price, less quantity of the commodity is offered for sale. There is a direct relationship between price and quantity supplied as shown by law of supply.

b) Price of Related Good (Z)

Supply of a commodity depends upon the prices of its related goods, specially substitute goods. If the price of a commodity remains constant and the price of its substitute good Z increases, the producers would prefer to produce substitute good Z. As a result, the supply of commodity X will decrease and that of substitute good Z will increase. This will shift the supply curve of good X leftward. Thus, an increase in the price of substitute good will lead to decrease in supply curve of the other good and vice-versa.

c) State of Technology

If there is a change in the technique of production leading to a fall in the cost of production, supply of commodity will increase

Examples.

New photostating technique, printing technique, computerised calculations, etc. Such advancement will lower the Marginal Cost (MC) at each level of output. It is shown in Fig. 2.1

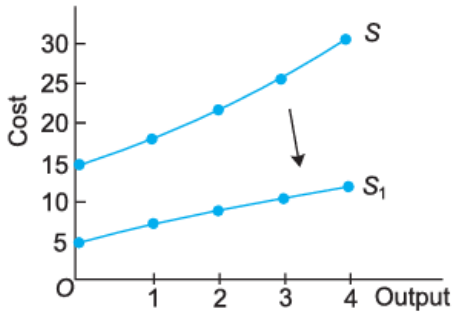


Fig. 2.1 Effect of Technological Progress on MC

Thus, when MC values are plotted, the new MC curve lies below the old MC curve. Rising portion of MC curve is the supply curve. Thus, with technological advancement supply curve shifts to the right (that is, supply will increase).

d) Prices of Inputs

A change in the cost of production, i.e., prices of factors of production also affects the supply of a commodity. If wages of labour or price of raw materials increase, then MC of production will rise. As a result, supply of the good will fall because producers would prefer to produce some other commodities that can be produced at a lower cost. It is shown in Fig. 2.2.

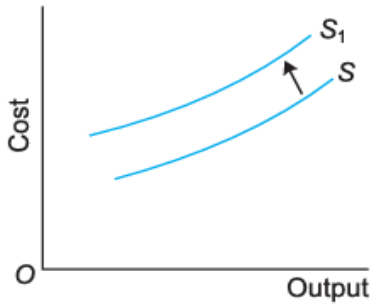


Fig. 2.2 Effect of Cost on MC Curve

The figure shows that increase in MC is shown by the new MC curve. Since MC curve is the supply curve there is a leftward shift in the supply curve. Thus, an increase in input price or cost will shift the supply curve to the left (decrease in supply) and vice-versa.

e) Government Policy

Government's policy also affects the supply of a commodity. If heavy excise taxes are imposed on a commodity, it will discourage producers and as a result, its supply will decrease. It is because excise duty is levied on the total production cost of a firm. An increase in excise duty will raise firm's total variable cost, which will raise MC curve. MC curve will shift to the left (The diagram will be same as Fig. 1.12.). Thus, supply curve will also shift to the left. Thus, an increase in excise tax will shift the supply curve to the left and vice-versa.

The Law of Supply

Law of supply derives the relationship between price and quantity supplied. According to the law of supply, other things remaining the same, quantity supplied of a commodity is directly related to the price of the commodity. In other words, other things remaining the same, when

price of a commodity rises, its quantity supplied increases and when the price falls, quantity supplied also falls. Symbolically, the law of supply is expressed as:

$$\mathbf{S_X = f\ (P_X)}, \text{ ceteris paribus}$$

For example, suppose a firm supplies 100 units of a good at price of ₹10 per unit. When its price rises to ₹15 per unit the firm supplies 150 units of the good and vice versa.

Assumptions of the Law of Supply:

The law of supply is based on the assumptions that all factors, other than the price of the commodity, that affect the supply remain the same. These are:

1. Prices of the related good should remain unchanged.
2. Prices of factors of production (i.e., prices of inputs) should remain unchanged.
3. Level of technology should remain unchanged.
4. Government policy regarding taxation should remain unchanged.
5. Goals of the firm should remain unchanged.

The Supply Schedule

Supply schedule is a tabular statement that gives the law of supply, i.e., it gives the different quantity supplied of a commodity at different prices per unit of time. A hypothetical supply schedule of wheat is given in Table 2.1.

Table 2.1 Supply Schedule of Wheat

Price (₹ per kg)	Quantity Supplied (kg per Month)
1	5
2	8
3	12

The supply schedule obeys the law of supply, i.e., as price of wheat rises, its supply also rises.

Supply Curve

Supply curve shows graphically the relationship between quantity supplied of a commodity to its price. The curve shows positive or direct relationship between the price and quantity

supplied of the commodity. With rise in price, the curve rises upward from left to the right as shown in Fig. 2.3.

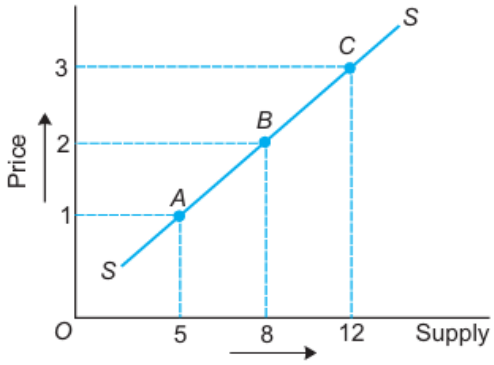


Fig. 2.3 The Supply Curve

SS is the upward sloping supply curve obeying the law of supply.

Reasons Behind Upward Sloping Supply Curve

The main reasons behind an upward sloping supply curve are:

(a) Law of Diminishing Marginal Productivity.

The law states that as more units of the variable factor are employed, the addition made to total production falls, i.e., cost of production rises. Thus, more quantity is supplied only at higher prices so as to cover the rise in cost of production.

(b) Goal of Profit Maximisation.

The aim of producers is to maximise profits. The aim can be achieved by raising the price of the goods. At higher price producers increase the supply of the goods.

Summing Supply Curves

a) Market Supply Schedule

An individual supply indicates different quantities offered for sale by an individual firm at different price in a particular time period. A market supply shows total quantity of the commodity offered for sale by all firms at different prices in a particular time period. Thus, market supply is obtained by aggregating the supplies of all firms selling that commodity at alternative prices. Suppose, there are only two firms A and B in the market for wheat. Market supply schedule is given in Table 2.2.

Table 2.2 Construction of Market Supply Schedule

Price (₹ per kg)	Individual Supply Schedules (kg per month)		Market Supply Schedule A + B (kg per month)
	Firm A	Firm B	
3	8	13	21
2	6	8	14
1	0	5	5

b) Market Supply Curve

Supply curve is a graphical representation of supply schedule. Individual supply curve reflects an individual supply schedule and market supply curve represents a market supply schedule. Market supply curve is obtained by horizontal summation of all individual supply curves as shown in Fig. 2. 4.

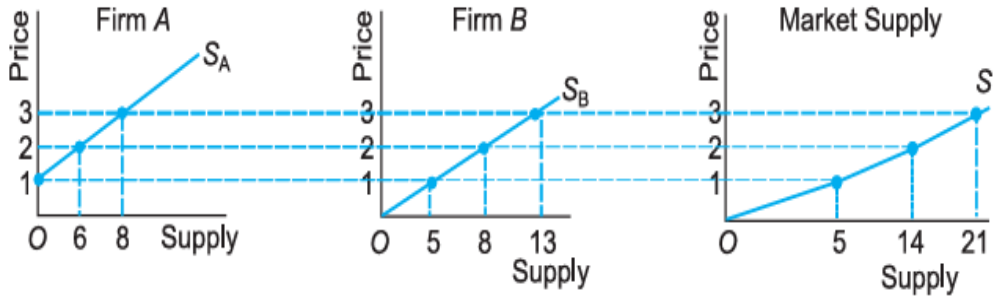


Fig. 2.4 Construction of Market Supply Curve

In the figure, quantity supplied is taken on the x-axis and price at which commodity is supplied on the y-axis. S_A and S_B are individual supply curves. SS is the market supply curve which is obtained by horizontally aggregating S_A and S_B at each level of price.

Change in Quantity Supplied (Movement)

A movement along the supply curve is caused by changes in the price of the good, other things remaining constant. It is also called change in quantity supplied of the commodity. In a movement, no new supply curve is drawn. Movement along a supply curve can bring about:

(a) Expansion or extension of supply, or

(b) Contraction of supply.

Expansion or extension of supply refers to rise in supply due to rise in price of the good. Contraction of supply refers to fall in supply due to fall in price of the good. Expansion or contraction of supply curve is shown with the help of original and revised supply schedules given in Table 1.9.

Table 2.3 Expansion or Contraction of Supply

Original Supply Schedule		**Revised Supply Schedule**						
Price (₹)	Quantity (Units)	Ref. Point	Expansion		Ref. Point	Contraction		Ref. Point
			P_X	Q_X		P_X	Q_X	
10	20	A	15	30	C	5	10	B

Movement along a supply curve is graphically shown in Fig. 2.5. Point A on the supply curve is the original situation.

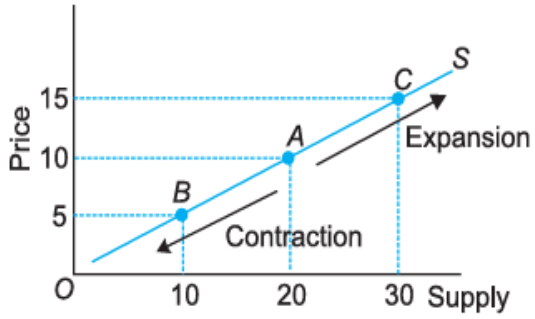


Fig. 2.5 Movement along Supply Curve

An upward movement from point A to a point such as C shows expansion or more supply at a higher price. A downward movement from point A to a point such as point B shows contraction or less supply at a lesser price.

Change in Supply (Shift)

A change (or shift) in supply curve is caused by changes in factors other than the price of the good. A change in many factors causes shift in the supply curve. It is also called change in supply. In a shift, a new supply curve is drawn. A shift of the supply curve can bring about:

(a) Increase in supply, or

(b) Decrease in supply.

(a) Increase in Supply (i.e., Rightward shift in supply curve)

When supply of a commodity rises due to favourable changes in factors other than price of the commodity, it is called increase in supply. Favourable changes imply:

(i) Improvement in technique of production

(ii) Fall in the price of related goods

(iii) Fall in the prices of inputs

(iv) Fall in excise tax

Increase in supply means more quantity supplied at the same price. It also means that same quantity supplied at a lower price. Increase in supply can be shown with the help of a supply schedule as given in Table 2.4.

Table 2.4 Increase in Supply

Original Supply Schedule		**Increase in Supply**	
P_x (₹)	Q_x (Units)	P_x (₹)	Q_x (Units)
10	20	10	30

Increase in supply is graphically shown in Fig. 2.6 where quantity supplied is measured on the x-axis and price of the commodity on the y-axis. SS is the original supply curve.

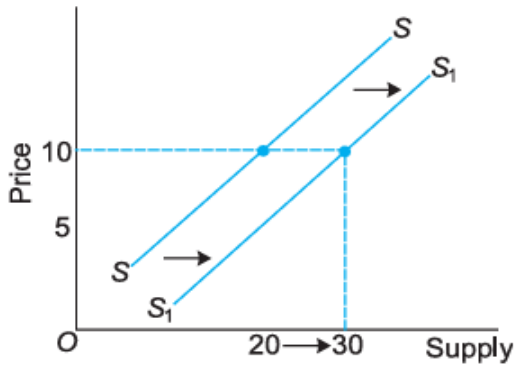


Fig. 2.6 Rightward Shift in Supply Curve: Increase in Supply

An increase in supply is shown by rightward shift of the supply curve from SS to S_1S_1. An increase in supply shows that:

(i) either at the original price of ₹10, more units (30 units) of the good are supplied. In the original situation 20 units were supplied.

(ii) or same units (20 units) are supplied at a lower price of ₹5.

(b) Decrease in Supply (i.e., Leftward shift in supply curve)

When supply of a commodity falls due to unfavourable changes in factors other than its price, it is called decrease in supply. The causes of decrease in supply are:

(i) Obsolete technique of production

(ii) Increase in the price of related goods

(iii) Increase in the prices of inputs

(iv) Rise in excise tax.

Decrease in supply means less quantity is supplied at the same price. It also means that same quantity is supplied at a higher price. Decrease in supply is shown with the help of a supply schedule in Table 2.5 and graphically in Fig. 2.7.

Table 2.5 Decrease in Supply

Original Supply Schedule		**Decrease in Supply**	
P_x (₹)	Q_x (Units)	P_x (₹)	Q_x (Units)
10	20	10	10

In the figure, SS is the original supply curve. A decrease in supply is shown by leftward shift of the supply curve from SS to S_1S_1.

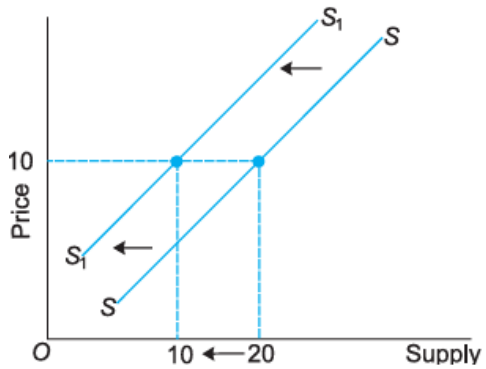


Fig. 2.7 Leftward Shift in Supply Curve: Decrease in Supply

A decrease in supply shows that:

(i) either at the original price of ₹10, lesser units (10 units) of the good are supplied. In the original situation 20 units were supplied.

(ii) or same units (20 units) are supplied at a higher price of ₹20. The difference in the causes of increase and decrease in the supply is summarised in Table 2.6.

Table 2.6 Difference in the Causes of Shift in the Supply

Increase in Supply (Rightward shift of supply)	**Decrease in Supply (Leftward shift of supply)**
1. Improvement in technique of production 2. Fall in the price of related goods 3. Fall in cost 4. Fall in excise tax	1. Technique of production becoming obsolete 2. Rise in price of related goods 3. Rise in cost 4. Rise in excise tax

Multiple Choice Questions

1. What is constant in the law of supply?

(a) Price of related goods (b) State of technology

(c) Cost of production (d) All of the above

Answer: (d)

2. When supply curve is upward sloping, its slope is _____.

(a) Positive (b) Negative

(c) First positive then, negative (d) Zero

Answer: (a)

3. Market supply curve is _____ summation of individual supply curves.

(a) Horizontal (b) Vertical

(c) Can be both horizontal or vertical (d) None of the above

Answer: (a)

4. Movement along the supply curve is also called:

(a) Change in supply (b) Change in quantity supplied

(c) Contraction in supply (d) Increase in supply

Answer: (b)

5. An upward movement along a supply curve shows:

(a) Contraction in supply (b) Decrease in supply

(c) Expansion in supply (d) Increase in supply

Answer: (c)

6. A rightward shift in supply curve shows:

(a) Contraction in supply (b) Decrease in supply

(c) Expansion in supply (d) Increase in supply

Answer: (d)

7. When less quantity is supplied at a lower price, it shows:

(a) Contraction in supply (b) Decrease in supply

(c) Expansion in supply (d) Increase in supply

Answer: (a)

8. When same quantity is supplied at a higher price, it shows:

(a) Contraction in supply (b) Decrease in supply

(c) Expansion in supply (d) Increase in supply

Answer: (b)